

```
In [1]: import sys
import platform

print("Python version  :", sys.version.split()[0])
print("Interpreter path :", sys.executable)
print("Platform         :", platform.platform())
print()
print("Running in ai-lab:", "ai-lab" in sys.executable)
```

```
Python version  : 3.11.16
Interpreter path : C:\Users\DELL\anaconda3\envs\ai-lab\python.exe
Platform         : Windows-10-10.0.26200-SP0
```

```
Running in ai-lab: True
```

```
In [9]: print("Hello, Introduction to Artificial Intelligence")
```

```
Hello, Introduction to Artificial Intelligence
```

```
In [3]: import pandas as pd
print("pandas version:", pd.__version__)
```

```
-----
ModuleNotFoundError                                Traceback (most recent call last)
Cell In[3], line 1
----> 1 import pandas as pd
      2 print("pandas version:", pd.__version__)

ModuleNotFoundError: No module named 'pandas'
```

```
In [5]: !pip install pandas
```

```
Requirement already satisfied: pandas in .\anaconda3\envs\ai-lab\Lib\site-packages (3.0.5)
Requirement already satisfied: numpy>=1.26.0 in .\anaconda3\envs\ai-lab\Lib\site-packages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in .\anaconda3\envs\ai-lab\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in .\anaconda3\envs\ai-lab\Lib\site-packages (from pandas) (2026.4)
Requirement already satisfied: six>=1.5 in .\anaconda3\envs\ai-lab\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
In [7]: import pandas as pd
print("pandas version:", pd.__version__)
```

```
pandas version: 3.0.5
```

```
In [10]: import pandas as pd

url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"
df = pd.read_csv(url)

print("Shape (rows, columns):", df.shape)
df.head()
```

Shape (rows, columns): (891, 15)

```
Out[10]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody
tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>survived</th> <th>pclass</th> <th>sex</th> <th>age</th> <th>sibsp</th>
<th>parch</th> <th>fare</th> <th>embarked</th> <th>class</th> <th>who</th>
<th>adult_male</th> <th>deck</th> <th>embark_town</th> <th>alive</th>
<th>alone</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> </tbody> <tbody></tbody>
```

0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
In [11]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    str
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    str
8   class       891 non-null    str
9   who         891 non-null    str
10  adult_male  891 non-null    bool
11  deck        203 non-null    str
12  embark_town 889 non-null    str
13  alive       891 non-null    str
14  alone       891 non-null    bool
dtypes: bool(2), float64(2), int64(4), str(7)
memory usage: 92.4 KB
```

```
In [12]: df.describe()
```

```
Out[12]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody
tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>survived</th> <th>pclass</th> <th>age</th> <th>sibsp</th>
<th>parch</th> <th>fare</th> </thead> <tbody> <th>count</th> <th>mean</th>
<th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th>
<th>max</th> </tbody> <tbody> </tbody>
```

891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [13]: missing = df.isnull().sum()
missing = missing[missing > 0].sort_values(ascending=False)

print("Columns containing missing values:\n")
print(missing)
print()
print("As a percentage of all rows:\n")
print((missing / len(df) * 100).round(1))
```

Columns containing missing values:

```
deck          688
age           177
embarked       2
embark_town    2
dtype: int64
```

As a percentage of all rows:

```
deck          77.2
age           19.9
embarked       0.2
embark_town    0.2
dtype: float64
```

```
In [ ]:
```